

Metabolism constrains the ecology of the peri-implant biofilm

genome-scale cross-feeding fixes the sign of interactions; dysbiosis is a re-wiring & spatial re-organisation — not a bulk shift

1 Defined community

5-species HOBIC biofilm (in vitro)
+ 10-guild in-vivo 16S (Dieckow)

 *S. oralis*

 *A. naeslundii*

 *V. dispar/parvula*

 *F. nucleatum*

 *P. gingivalis*

titanium implant surface

2 The principle

AGORA2 genome-scale FBA

who secretes / consumes which metabolite

j secretes X , i consumes X

$\Rightarrow$ **cross-feeding** $\Rightarrow A_{ij} > 0$

j secretes a toxin (H_2O_2 , H_2S)

$\Rightarrow A_{ij} < 0$

$\text{sign}(A)$ from metabolism · magnitude from data

3 Commensal $\rightleftharpoons$ dysbiotic

COMMENSAL

 $\rightleftharpoons$ 
 $So \rightleftharpoons Vd$ mutualism

shallow,
balanced

X

DYSBIOTIC

 $\not\rightleftharpoons$   Fg central & deep
mutualism breaks

validated: cross-feeding direction $p = 4 \times 10^{-4}$ · two cohorts (Dieckow $\times$ Botelho) 89% prior-free · mechanistic COMETS dFBA reproduces the split